

JPEG AI as a Threat to Deepfake Detection

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Problem Statement

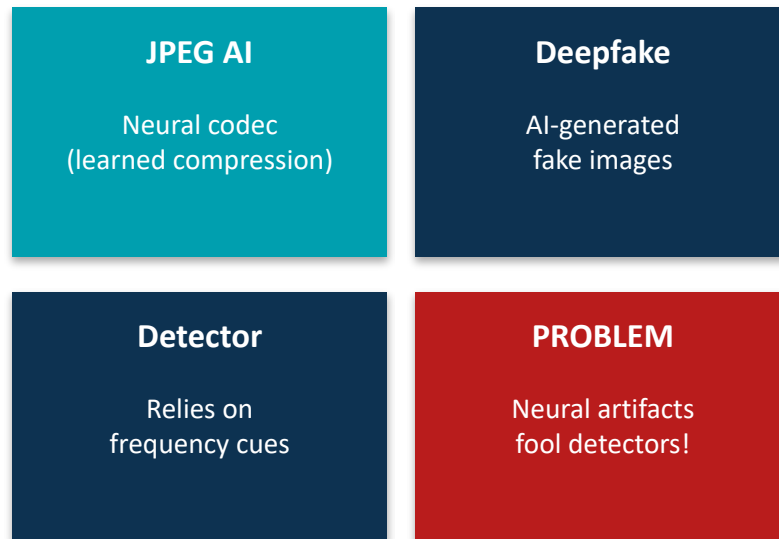
JPEG AI: A New Compression Paradigm

- First international learned image compression standard (ISO/IEC)
- Deep neural networks used for encoding and decoding
- Artifacts with fundamentally different statistical signature than classical JPEG
- Increasingly adopted — threatens existing forensic pipelines

The Forensic Challenge

Deepfake detectors exploit low-level frequency cues and compression artifacts from the generation process.

JPEG AI disrupts these forensic fingerprints, dramatically degrading detector performance.



↓ JPEG AI recompression ↓

Research question:
*How much does JPEG AI compression degrade
deepfake detection performance?*

State of the Art

Deepfake Detection

CLIP-Det (Cozzolino et al., CVPRW 2024): CLIP features + linear classifier.
Corvi2023: ResNet50 fine-tuned on diffusion-generated images.
Detectors exploit low-level statistical fingerprints in the image spectrum.

Classical Compression (JPEG)

Traditional JPEG introduces well-studied block-DCT artifacts.
Detectors degrade gracefully; retraining on JPEG-augmented data restores robustness.
JPEG robustness is a standard forensic benchmark.

JPEG AI & Forensics

Cannas et al. (ICCVW 2025): first study on JPEG AI impact on image forensics.
Neural codecs produce periodicity artifacts from upsampling layers.
These differ fundamentally from block-DCT artifacts => detectors fail unexpectedly.

Mitigation Approaches in Forensics

Preprocessing (filtering) to suppress codec-specific artifacts.
Domain adaptation / fine-tuning on compression-augmented data.
Score fusion to combine complementary detector outputs.

Proposed Method — Three-Phase Study

Phase 1

Baseline Evaluation

- Apply JPEG AI compression at BPP levels 0.12 – 1.00
- Evaluate CLIP-Det (3 score variants) on original vs. compressed images
- Measure AUC / Accuracy / F1 as a function of bitrate

Phase 2

Frequency-Domain Analysis

- 2D Fourier spectra of noise residuals
- 1D radial power profiles: real vs fake
- DCT coefficient distributions under compression
- Spectral distance vs AUC drop correlation

Phase 3

Mitigation Strategies

- Preprocessing (median, Gaussian, JPEG-75)
- Per-BPP logistic regression score fusion (calibration-based)
- Linear-probe fine-tuning of the Corvi2023 detection head

Dataset

OpenFake 20k

Balanced dataset of real and AI-generated face images, widely used as a deepfake detection benchmark.

Total images	~20,000
Classes	Real / Fake (1:1)
Resolution	Variable (RGB)
Source	Kaggle (AI-generated faces)
Test sample	200 images: 100 real + 100 fake

JPEG AI Compression Pipeline

- 1 Convert all images to PNG for codec compatibility
- 2 Compress with JPEG AI reference software (GitLab)
- 3 4 target BPP levels: 0.12 · 0.25 · 0.50 · 1.00
- 4 Reconstruct and verify all output files
- 5 Build comp_df manifest with measured BPP metadata

Target BPP levels:



Experimental Setup

Detector: CLIP-Det

ClipBased-SyntheticImageDetection (GRIP-UNINA)

CLIP features + linear classifier · 3 output scores:
clipdet_latent10k_plus, Corvi2023, fusion

Primary score: Corvi2023 (AUC > 0.75 on uncompressed data)

Framework & Environment

Language: Python

Frameworks: Pytorch, JPEG AI

Environment: Google Colab

Evaluation Metrics

Primary: **AUC** — threshold-independent, reliable

Secondary: Accuracy, F1-score (threshold: LLR > 0)
Reported per BPP level and per detector variant

Calibration / Test Split (Phase 3)

50% calibration / 50% test split (stratified, seed=42)

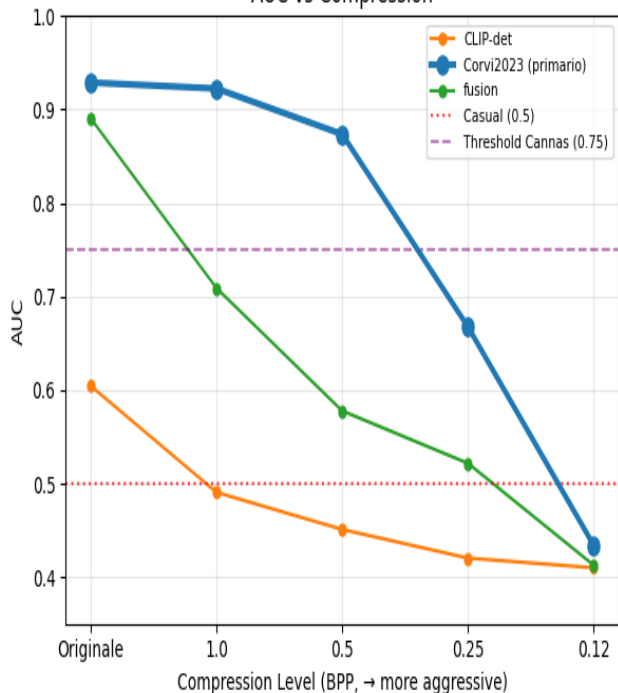
Calibration: used to fit Strategies B & C

Test: held-out evaluation — no data leakage

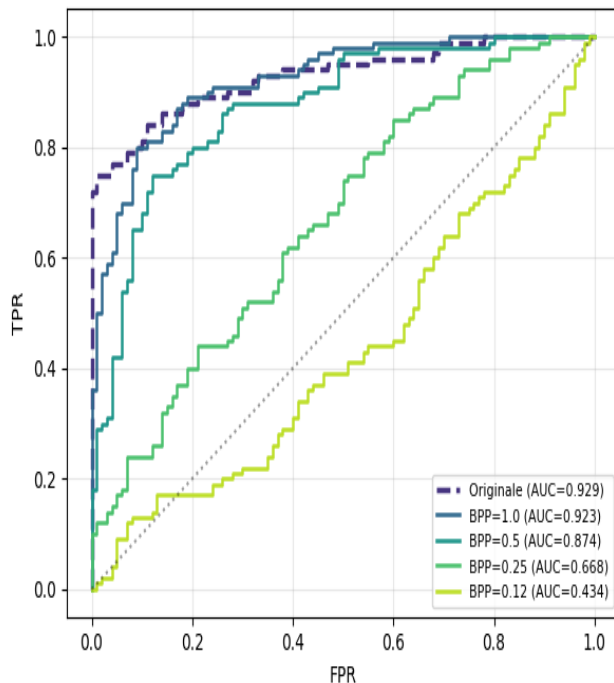
Phase 1 — Baseline Evaluation under Compression

Phase 1 — JPEG AI as threat to deepfake detection

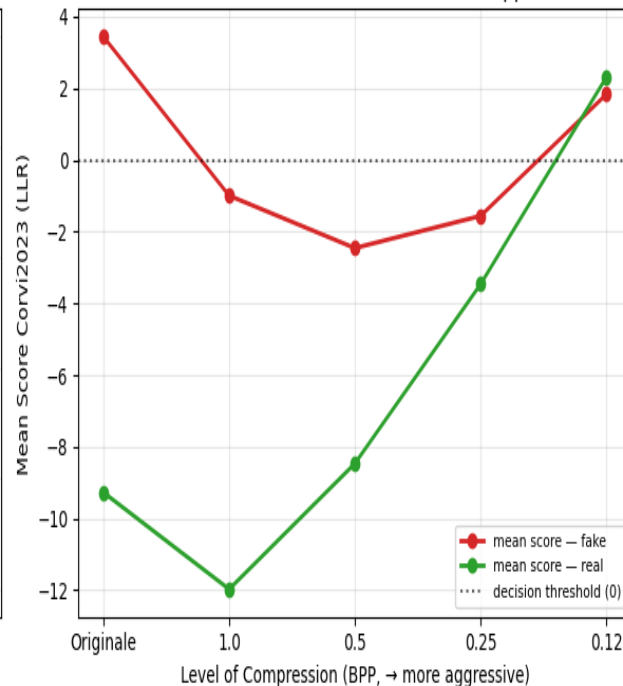
AUC vs Compression



Curves ROC — Corvi2023

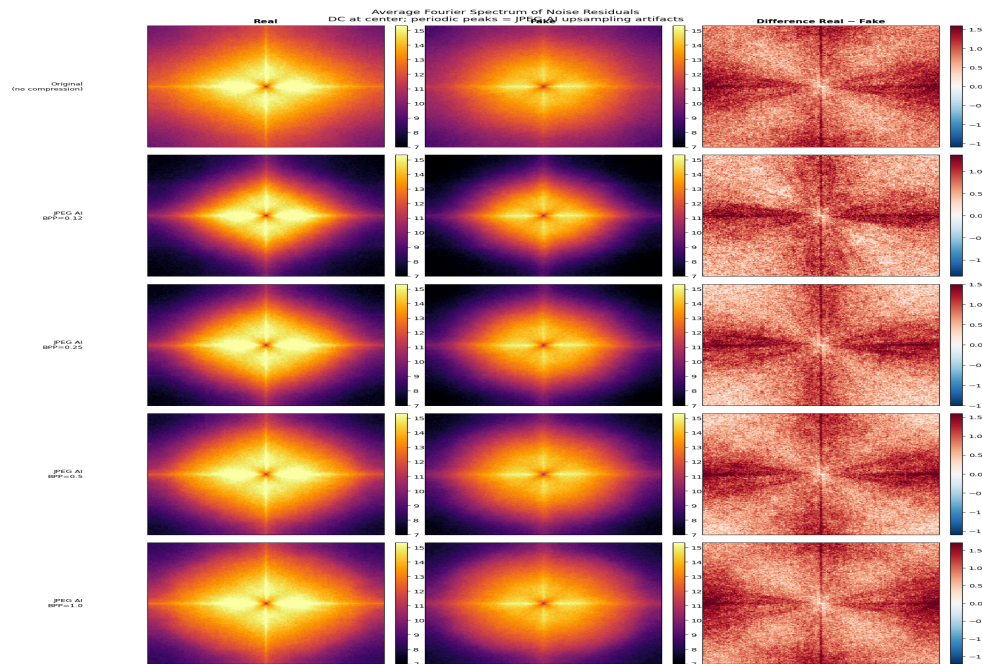


Inversion: at low BPP real/fake are swapped



Key findings: AUC drops sharply at low BPP · Counter-forensic inversion at BPP=0.12 · Corvi2023 is the most resilient detector

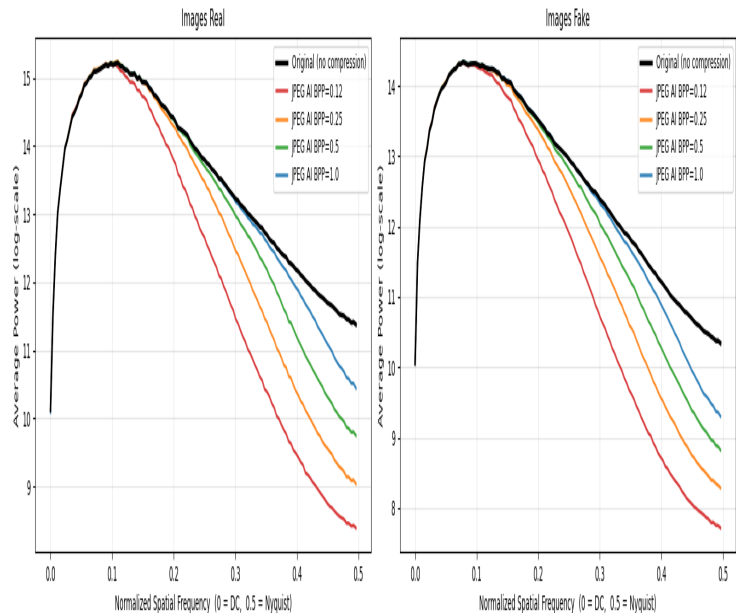
Phase 2 — Frequency Analysis: Fourier Spectra



2D Average Power Spectrum

JPEG AI introduces periodic peaks from upsampling.
These artifacts are absent in original images.

Radial Profile of the Power Spectrum of Noise Residuals
(peaks = periodicity of the JPEG AI decoder)

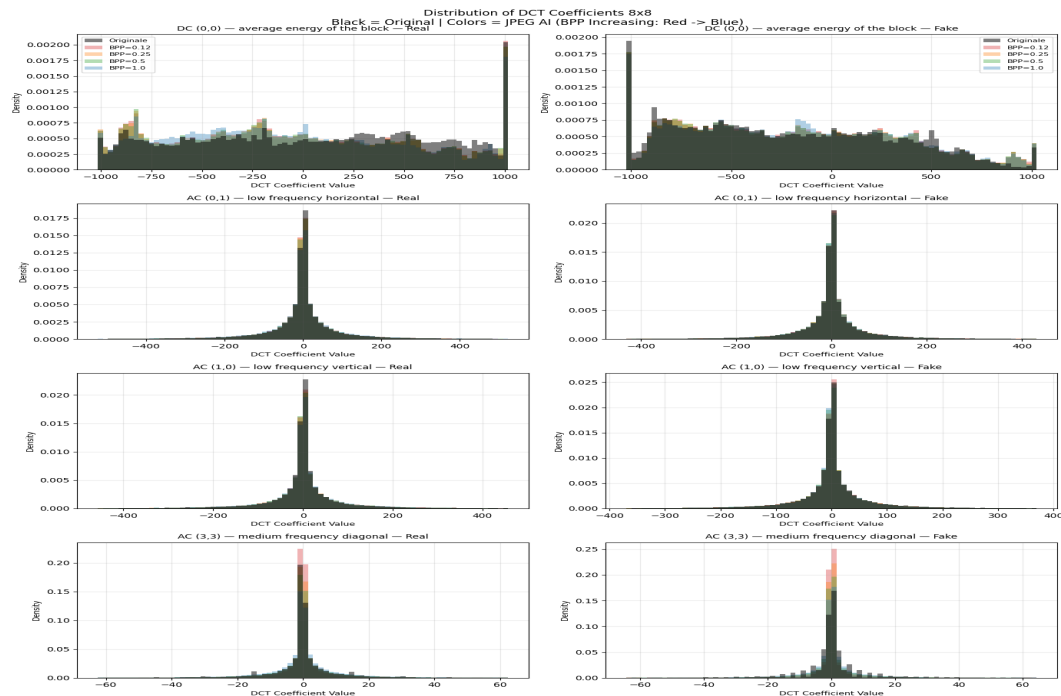


Radial Power Profile

Real and fake spectral profiles converge at low BPP.
The detector discriminative signal collapses.

Phase 2 — DCT Coefficient Distributions

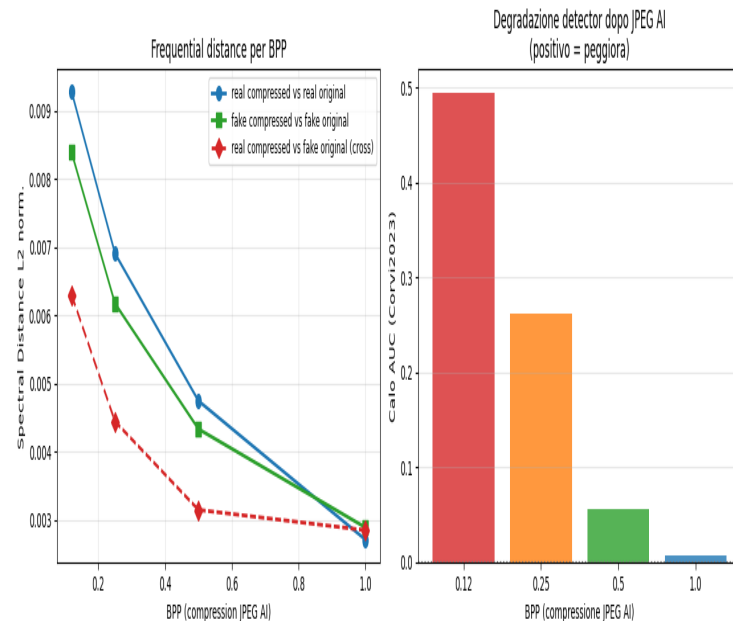
& Spectral Distance vs. AUC Drop



8x8 DCT Coefficient Distributions

JPEG AI smooths high-frequency DCT coefficients differently from JPEG, erasing forensic cues detectors rely on.

Spectral Distance Introduced by JPEG AI and Detector Degradation



Spectral Distance vs. AUC Drop

Strong correlation: higher spectral perturbation predicts larger AUC degradation.

Phase 3 — Mitigation Strategies

Strategy A

Compression-Aware Preprocessing

Training-free · Applied before inference

median3: 3×3 median filter

gaussian: Gaussian blur ($\sigma = 0.8$)

jpeg75: JPEG re-encoding at quality 75

Aim: suppress JPEG AI neural fingerprints while preserving the detection signal.

Strategy B

Per-BPP Score Fusion (Logistic Regression)

Calibration-based · Low computational overhead

CLIP-Det outputs 3 complementary scores.

A logistic regression fitted per BPP level on calibration scores re-weights each detector.

Aim: restore discriminative power when one detector inverts under compression.

Strategy C

Linear-Probe Fine-Tuning

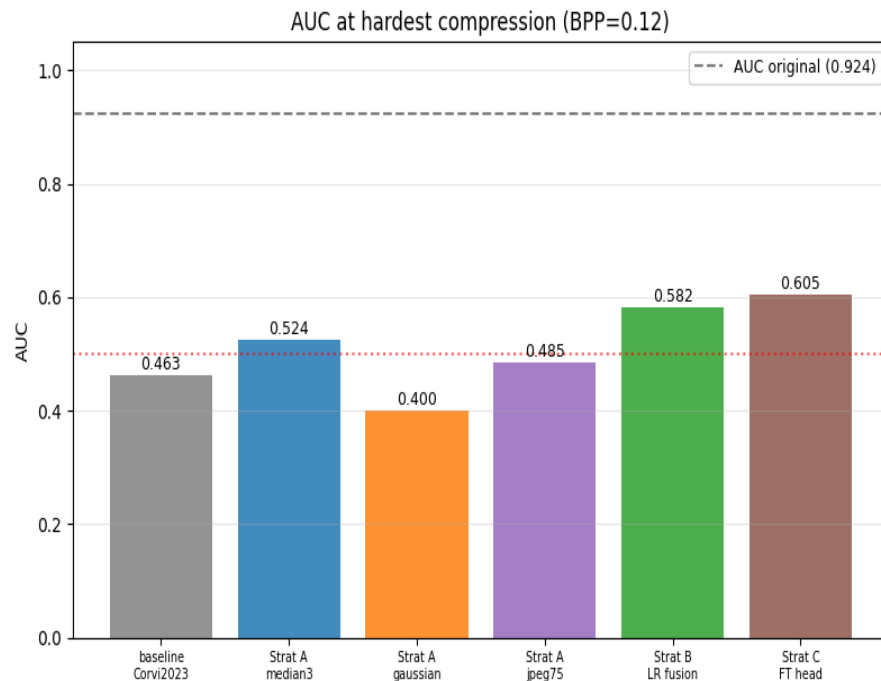
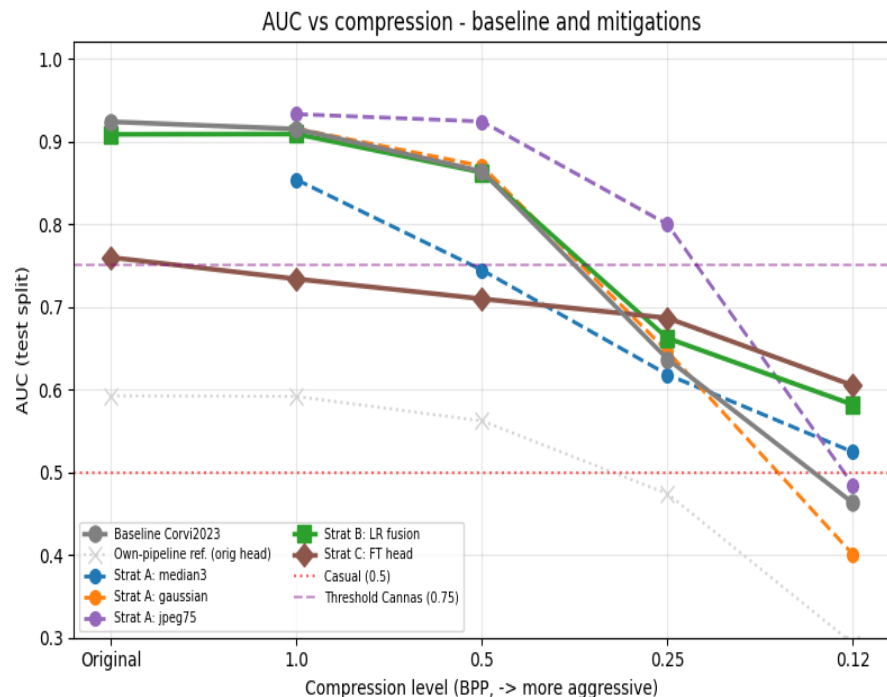
Calibration-based · Zero extra inference cost

ResNet50 backbone (Corvi2023) frozen. Only the Linear(2048→1) head retrained on multi-BPP augmented calibration data.

Aim: new decision boundary robust to JPEG AI feature distribution shift.

Phase 3 — Mitigation Results

Phase 3 - Mitigation strategies (primary detector: Corvi2023)



Main findings: Strategies B & C recover significant AUC at low BPP · Strategy A provides modest gains with zero training cost · Trade-off: effectiveness vs. computational overhead

Conclusions

1

JPEG AI significantly degrades deepfake detection

AUC drops from > 0.75 to below 0.5 at BPP = 0.12 — counter-forensic inversion.

2

Frequency analysis explains the mechanism

JPEG AI upsampling artifacts eliminate the spectral fingerprints that detectors rely on.

3

Mitigation is feasible

Strategies B & C recover 10–25% AUC at low BPP. Strategy A is competitive with zero training.

References

- [1] E. D. Cannas et al., "Is JPEG AI Going to Change Image Forensics?" ICCVW 2025, pp. 1575-1586.